A Comprehensive Analysis of LayoutLMv3 and RoBERTa for Classification and Key Information Extraction Fine-Tuned on the DocILE Dataset.

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Outline

Document AI (DocAI)

- **DocAI** is the process of understanding, classifying, and extracting information from visually rich documents (VRDs) of different typesetting formats such as HTML/XML web pages, digitally born documents and scanned documents (Cui et al.).
- VRDs, such as forms, receipts, and invoices, are widely present in various businesses.
- Due to the diversity of layouts and formats, low-quality scanned document images, and the complexity of document template structures, DocAI poses a very challenging task (Cui et al).

Introduction

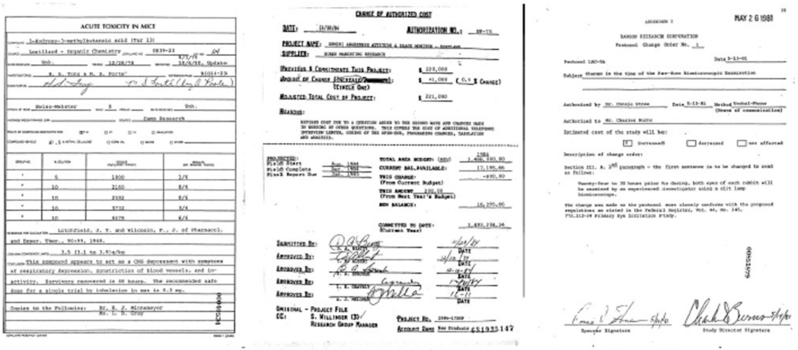
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Figure: 1. Scanned images of business documents with different layouts and formats (Yiheng Xu et al.)

Motivation

- Manual data entry is a labor-intensive and time-consuming process (Mehmet Selman Baysan et al.).
- Therefore, automating the extraction and classification of information saves human effort, improve efficiency, productivity, and accuracy

Objectives

1. Evaluate performance of state-of-the-art models (RoBERTa and LayoutLM) on the DocILE Dataset for Key Information Localization and Extraction (KILE).
2. Implement fine-tuning strategies on SOTA models to investigate their impact on performance.
3. Compare baseline models against fine-tuned models to evaluate improvements in accuracy, generalization, and efficiency.

Timeline

- **Rule based heuristic**
  - Manually summarizing these rules incurred huge labor costs and was not scalable
- **Machine Learning Methods**
  - Can automatically learn patterns from data, reducing manual effort and improving scalability, but they require large annotated data.
- **Deep Learning Methods**
  - These methods automatically learn complex features and hierarchical representations from raw data.

Document AI Tasks

1. **Document Layout Analysis (DLA)**
   - Automatic identification, recognition, and interpretation of text, images, tables, figures and charts
2. **Visual Information Extraction (VIE)**
   - Extracting entities and their relationships from unstructured documents.
3. **Document Visual Question Answering (DocVQA)**
   - Ability of the system to answer questions about digital-born documents
4. **Document Image Classification (DIC)**
   - Classifying documents into categories, such as resumes, invoices, receipts

DocAI Pipeline

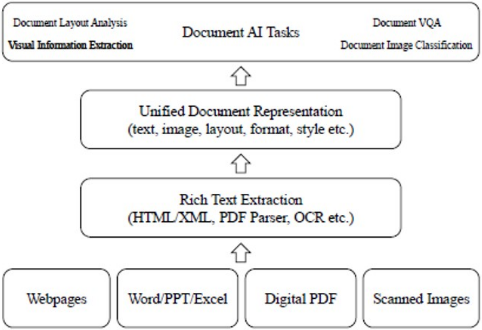

Figure: 2. Overview of Document AI (Cui et al).

DocAI Pipeline

- **Optical Character Recognition (OCR)**
  - Extract text and bounding boxes
- **Convolutional Neural Networks (CNN)**
  - Extracting grid data such as images
- **Graph Neural Networks (GNN)**
  - Extracting data organized in graph structures.

BERT (Devlin et al.)

- Bidirectional Encoder Representations from Transformers
- Text-only transformer-based model proposed by Devlin et al.
- Captures contextual relationships between words/tokens as they are purely designed for text input
- Pretrained on two unsupervised tasks:
  - Masked Language Modeling (MLM): random tokens are masked and the model is trained to predict them
  - Next Sentence Prediction (NSP): Trains the model to predict whether two segments naturally follow each other.
- However, BERT is significantly undertrained. (16GB of uncompressed text).

RoBERTa (Liu et al.)

- Robustly Optimized BERT
- Built upon BERT with the following pre-training modifications:
  - Model is trained longer.
  - The NSP objective is removed
  - Training is performed on longer sequences
  - The masking pattern is dynamically applied to the training data
- Both BERT and RoBERTa excel at text understanding but fail to capture spatial layout and visual cues present in documents

RoBERTa

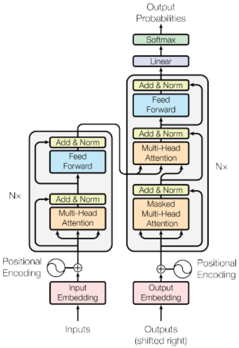

Figure: 3. Transformer Architecture (Vaswani et al.)

Limitations

- VRDs contain various visual information like font types, size, color.
- These visual cues carry strong semantic signals
- Therefore, to address the limitations of text-only transformer-based model, several multimodal models have been developed that incorporate textual, layout and visual embeddings.

Proposed Multimodal Models

- DocFormer:
  - A multimodal transformer-based architecture which combines text, vision and spatial features
- FormNet:
  - A structure aware sequence model, that achieves state-of-the-art performance on form-like documents.
- UDOP:
  - A unified multimodal pretraining framework that jointly learns from text, layout, and images to support a wide range of document understanding tasks, from captioning to extraction.

LayoutLM (Yiheng Xu et al.)

- LayoutLM extends BERT by incorporating both text(tokens) and layout(bounding boxes).
- Pretrained on two self-supervised tasks:
  - **Masked Visual-Language Model (MVLM):**
    - The model learns language representation from bounding boxes and tokens by masking some input tokens and predicting them given their bounding boxes.
  - **Multi-Label Document Classification**
    - the model uses document tags, given a set of scanned documents, to generate high-quality document-level representations.
- However, LayoutLM does not incorporate visual information during the pre-training process but rather combined in the fine-tuning stage.

LayoutLMv2 (Yang Xu et al.)

- Visual embeddings are integrated in the pre-training stage.
- Pre-training objectives:
  - **Text Image Alignment (TIA)**
  - **Text-Image Matching (TIM)**
- LayoutLMv2 significantly outperforms LayoutLM and achieves state-of-the-art performance on a variety of downstream DocAI tasks (Xu et al.).
- However, LayoutLMv2 uses CNN-based approach to enhance multimodal understanding of documents. This method results in high computational cost.

LayoutLMv3 (Huang et al.)

- First model in Document AI that doesn't rely on CNN to extract image features.
- LayoutLMv3 replaces CNNs with ViT-style patch embeddings by splitting the document image into fixed-size patches, flattening them, projecting them through a linear layer, and treating each patch as a visual token.
- Hence, enabling more efficient multimodal integration of text, layout, and visual information.

LayoutLMv3 (Huang et al.)

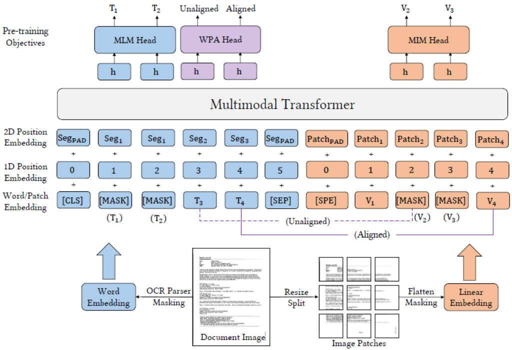

Figure: 3. LayoutLMv3 Architecture (Huang et al)

Benchmark Datasets

- It is crucial that high-quality benchmark datasets are available to evaluate and compare the performances of models.
- Existing benchmark datasets don't account for generalizing to unseen document templates, complex target schema, hierarchical entities, and small training datasets.
- A benchmark dataset should involve rich schema, layout-rich documents, diverse templates, high-quality OCR results, and token-level annotation (Z. Wang et al).

DocILE

- Document Information Localization and Extraction
    - DocILE contains business and form-style documents such as Invoices, Receipts, and Financial statements.
    - DocILE contains approximately 6,700 human-annotated documents, more than 100,000 synthetically generated documents and about a million unlabeled documents.
    - This makes it one of the most comprehensive publicly available resources for the tasks of Key Information Localization and Extraction (KILE) and Line Item Recognition (LIE).

Methodology I

1. **Dataset Preprocessing**
   - 6680 Annotated Documents
   - Train $\sim$ 5,180
   - Val $\sim$ 500
   - Test $\sim$ 1000
2. **Experimental Setup**
   1. Full Fine-Tuning
      - All layers are trained
      - Default parameters are retained
   2. Partial Fine Tuning
      - Selected portion of the model is trained (freeze layers)
      - Certain hyper-parameters are modified (learning rate, batch size, weight decay)

Methodology II

3. **Performance Evaluation:**
   - Quantitatively assess RoBERTa and LayoutLMv3 performance using metrics (accuracy, precision, recall, and F1-score) to establish baseline performance.
   - Use these baseline performance to compare against fine tuned RoBERTa and LayoutLMv3 models.

Results

- To evaluate the performance of RoBERTa and LayoutLMv3 on the DocILE dataset, a series of experiments such as full fine tuning and various partial fine tunings were conducted.

Results: RoBERTa

- Zero-Shot

Table: 1. Performance Metrics for Zero-Shot

Metric	Accuracy	F1	Recall	Precision
Value	0.0771	0.0771	0.0771	0.0771

- Full Fine Tuning

Table: 2. Performance Metrics for Full Fine Tuning

Metric	Accuracy	F1	Recall	Precision
Value	0.752303	0.588944	0.584045	0.628355

Results: RoBERTa

Table: 3. Partial Fine Tuning Strategies

	S1	S2	S3	S4	S5	S6
Freeze	8	6	6	4	3	4
Train	4	6	6	8	9	8
Top Layer LR	$1e^{-5}$	$3e^{-5}$	$3e^{-5}$	$4e^{-5}$	$5e^{-5}$	$5e^{-5}$
Classifier Head LR	$5e^{-5}$	$5e^{-5}$	$5e^{-5}$	$8e^{-5}$	$9e^{-5}$	$9e^{-5}$
Train Batch	16	16	16	24	24	24
Validation Batch	32	32	64	64	64	64
Epochs	5	5	5	5	5	5
Weight Decay	0.01	0.01	0.05	0.1	0.15	0.1
Warm Up Rate	5%	1%	1%	6%	8%	1%
Gradient Clipping	-	1.0	-	-	-	-
Scheduler	-	-	cosine	cosine	cosine with restarts	cosine
Adapter	-	-	-	-	-	LoRA

Results: RoBERTa

Table: 4. Performance Metrics for Partial Fine Tuning

	Accuracy	F1	Recall	Precision	F1 Micro
S1	0.736267	0.540493	0.544287	0.601609	0.736267
S2	0.748891	0.567848	0.571153	0.629968	0.748891
S3	0.736950	0.497926	0.483884	0.564740	0.736950
S4	0.747697	0.539008	0.543796	0.597693	0.747697
S5	0.744115	0.535737	0.545146	0.592028	0.744115
S6	0.693108	0.487520	0.453115	0.525531	0.693108

Results: RoBERTa

- Zero-shot demonstrated extremely poor performance, with an accuracy, F1-score, recall and precision all at 0.0771.
- Full fine-tuning achieved significant performance with an accuracy of 0.752303, F1 score of 0.588944, recall of 0.584045, and precision of 0.628355.
- Strategy S2 achieved the highest overall performance with an accuracy of 0.7489 and F1-score of 0.5678, closely approaching the full fine-tuned results.
- Strategy S6 performed weakest, despite incorporating LoRA adapters.

Results: LayoutLMv3

- Few-Shot

Table: 5. Performance Metrics for Few-Shot

Metric	Accuracy	F1	Recall	Precision
Value	0.52822766	0.4700191	0.500242822	0.443239425

- Full Fine Tuning

Table: 6. Performance Metrics for Full Fine Tuning

Metric	Accuracy	F1	Recall	Precision
Value	0.75601381	0.70472648	0.724114514	0.686349604

Results: LayoutLMv3

Table: 7. Performance Metrics for Partial Fine Tuning

	Accuracy	F1	Recall	Precision	F1 Micro
S1	0.620623	0.525354	0.492367	0.563079	0.607053
S2	0.756455	0.697855	0.711681	0.684557	0.715382
S3	0.705553	0.671339	0.691506	0.652316	0.686993
S4	0.701724	0.626102	0.643039	0.610035	0.668607
S5	0.624266	0.574169	0.526267	0.631665	0.603210
S6	0.671520	0.661920	0.671446	0.652660	0.658915

Results: LayoutLMv3

- The initial few-shot evaluation achieved an accuracy of 0.5282 and F1-score of 0.4700, significantly higher than RoBERTa's zero-shot performance.
- Full fine-tuning further improved the performance, achieving an accuracy of 0.7560 and F1-score of 0.7047, substantially outperforming RoBERTa full fine-tuning (0.5889 F1).
- Strategy S2 achieved the best performance, with an accuracy of 0.7564 and F1-score of 0.6979, closely matching the results achieved by full fine-tuning

Conclusion

- RoBERTa performed poorly in zero-shot highlighting that this model can't generalize well to unseen documents.
- LayoutLMv3 gained a higher performance over RoBERTa during full fine-tuning. Thus, underlining the importance of multimodal models.
- Overall, the results from both zero-shot and few-shot experiments highlight that there is still considerable progress to be made before we have models that can generalize reliably across diverse document types and datasets.

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THANK YOU!